We have checked that this model shows strong agreement with the data down to 100 keV.

To better understand the morphology of the IA signals from different FIP models, we compare them in Fig. 6. Here, we show in blue, the case of a sterile neutrino with coupling to τ particles $|U_{\tau-\nu_4}|^2 = 10^{-11}$, in green a DP with coupling $\epsilon = 2.5 \times 10^{-11}$, in yellow an ALP with $g_{ae} = 3 \times 10^{-18}$ and $g_{ap} = 2 \times 10^{-11}$ and, in maroon, a general FIP particle (described by Eq. (1)) injecting 5.5×10^{54} positrons per SN. These coupling values have been chosen to normalize them close to data in the region of COMPTEL measurements ($|l| < 60°$ and $|b| < 10°$) and refer to a 10 MeV mass. In addition, for comparison, we include the signal from a source of monoenergetic positrons (with energy of 10 MeV), as derived in Ref. [115]. This allows us to illustrate why the IA emission can be so constraining for FIPs: given that the energy of the positrons from these FIPs is always above a few tens of MeV, their IA emission is still very high up to these energies, where the γ-ray backgrounds at around 10-50 MeV are very low, mainly produced from IC emission of CR electrons. In fact, the high IA emission from high-energy positrons (above a few MeV) served to exclude heavy dark matter particles as the origin of the anomalous 511 keV line emission in Ref. [115] (but see also Refs. [111, 112]). Furthermore, positrons from FIPs produced in SNe have energies close to $\sim$ 100 MeV, where the cross sections for this process is more efficient. In addition, at these energies, the positrons still propagate for very long times, increasing the probability of interacting with electrons in the ISM.

To estimate the effect of uncertainties in propagation, we adopt the same strategy as used in Refs. [94, 100], where we adopted two extreme diffusion scenarios, a more aggressive one that leads to stronger constraints and a very conservative one that leads to weaker constraints. As we showed in Refs. [25, 94], the propagation parameters with a greater effect on the diffuse spectra produced from FIPs are the Alfvén velocity (V_A parameter) that controls the level of diffuse reacceleration [120–123] and the height (H) of the halo, which dictates the volume where CRs are confined and where FIPs produce electrons and positrons that can reach us. In the conservative setup, we set $H = 3$ kpc and $V_A = 0$ km/s (i.e. no reacceleration), which produces a lower (and therefore more conservative) flux of $e^{\pm}$. In turn, the more aggressive setup is meant to increase the flux of $e^{\pm}$ from the decay of FIPs, and uses values of $H = 16$ kpc and $V_A = 40$ km/s. These are at the extreme values within which we can still reproduce the current CR measurements, although they are not statistically favored [101, 103, 118, 124] [1] For comparison, we recall that our benchmark values are $H = 8$ kpc and

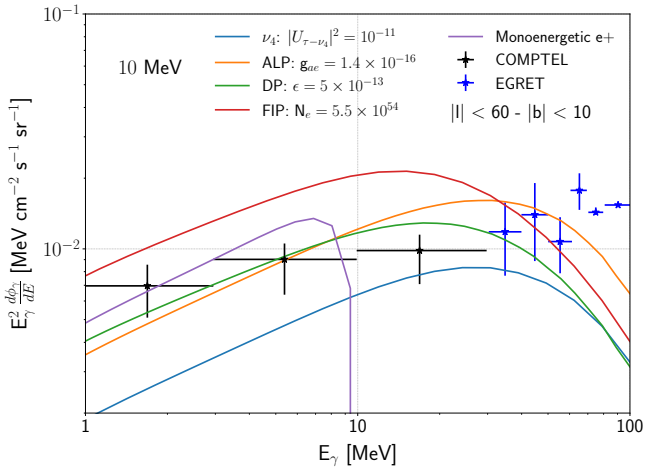

FIG. 6. Comparison of the morphology of the IA signals produced from different FIP models, in the region at $|l| < 60°$ and $|b| < 10°$, and for a mass of 10 MeV. We show in blue, the case of a sterile neutrino with coupling to τ particles (for $|U_{\tau-\nu_4}|^2 = 10^{-11}$), in green a DP with coupling $\epsilon = 2.5 \times 10^{-11}$, in yellow an ALP with $g_{ae} = 3 \times 10^{-18}$ and $g_{ap} = 2 \times 10^{-11}$ and in maroon, a general FIP particle (described by Eq. (1)) injecting 5.5×10^{54} positrons per SN. In addition, for comparison, we include the signal from a source of monoenergetic positrons (with energy of 10 MeV).

$V_A = 13.4$ km/s. In the left panel of Fig. 7, we show the uncertainty in the prediction of the IA emission from ALPs couplings to baryons and electrons for different masses. Each band represents the uncertainty in our estimated flux directly related to uncertainties in the propagation of CRs. As we will discuss later, these uncertainties translate into a factor of a few uncertainty in the constraint of the FIP coupling. We observe similar uncertainties for all the FIP models that we are studying in this paper.

Analysis of FIP γ-ray signals and constraints

As mentioned above, we derive 2σ constraints on the coupling for the different FIP models by comparing their IA emission with the measured Galactic γ-ray diffuse emission measured by COMPTEL in the $|l| < 60°$ and $|b| < 10°$ region of the sky. This direct comparison gives us a first very conservative constraint. However, on top of the positron-induced γ-ray continuum signals, the Galactic IC and bremsstrahlung emission from CR electrons are expected to be the dominant source of γ-ray production below energies of a few hundreds of MeV. Therefore, we also derive more realistic constraints for different estimations of the background model. In a first case, we

[1] We consider that $V_A = 40$ km/s is the maximum realistic value for V_A, since it already implies that most of the injected energy of CRs are coming from the perturbations of the interstellar plasma and not from supernova remnants, which will break the standard paradigm of CR propagation, see Ref. [121, 125]